Lecture 8

The symplectic form on solutions

Lecture 7 describes a field by its displacement and momentum on one time slice. A solution contains these data on every slice. We show that their antisymmetric pairing is independent of a compact deformation of the slice. The boundary term in the first variation determines its sign.

Objectives. *We construct the Klein–Gordon symplectic current, prove conservation of its integral on spacelike graphs, and prove nondegeneracy using compact Cauchy data. A finite spatial interval shows why side flux must be retained.*

By the end of this lecture you will be able to:

- Derive the symplectic current from the first-variation boundary term.
- Compute its integral on a spacelike graph with the future orientation.
- Prove slice independence and nondegeneracy for compact Cauchy data.

8.1 Solutions and their variations

Fix $m \geq 0$ on $\mathbb{R}^{1+d}$. The Klein–Gordon equation is $(\Box + m^2)\phi = 0$.

Definition 8.1 (Space of solutions)**.** The *space of solutions* $\mathcal{S}_m$ is the real vector space

$$\mathcal{S}_m = \{\phi \in C^\infty(\mathbb{R}^{1+d}, \mathbb{R}) : (\Box + m^2)\phi = 0, \quad (\phi(0), \phi_t(0)) \in C_c^\infty(\mathbb{R}^d)^2\}.$$

A *linearised solution* at ϕ is the derivative h of a smooth family of solutions through ϕ. Here the allowed variations are $h \in \mathcal{S}_m$, realised by the family $\phi + \varepsilon h$.

Linearity makes the linearised equation identical to the original equation: $(\Box + m^2)h = 0$. Theorem 4.3 constructs the solution of every compact smooth Cauchy datum. Such data have rapidly decreasing Fourier transforms, so the multiplier formula can be differentiated to every order. The resulting solution is smooth.

Theorem 5.3 gives a common compact spatial support on each bounded time interval. More precisely, if a datum is supported in $B_R(0)$, its solution vanishes outside $|x| \leq R + |t|$. Derivatives also vanish outside this set. These facts justify the integrals below.

We write δ for differentiation along solution families and d for spacetime exterior differentiation. For example, $\delta\phi(h) = h$ and $\delta p^\mu(h) = \partial^\mu h$, since $p^\mu = \partial^\mu \phi$ for the scalar field. These two differentiations commute because they act on separate variables. All field-space derivatives here can be evaluated on finite-dimensional affine families of solutions.

8.2 The boundary term and its antisymmetric variation

Recall $\mathrm{d}\Sigma_\mu = \iota_{\partial_\mu}\mathrm{vol}$ from Lecture 1. For any vector field A, the identity $d(A^\mu \mathrm{d}\Sigma_\mu) = (\partial_\mu A^\mu)\mathrm{vol}$ follows by differentiating its coefficients.

Definition 8.2 (Presymplectic potential and symplectic current)**.** The *presymplectic potential* is the spacetime d-form, linear in a variation h, given by

$$\theta_\phi(h) = p^\mu h \,\mathrm{d}\Sigma_\mu, \qquad p^\mu = \partial^\mu \phi. \tag{8.1}$$

The *symplectic current* is $\omega = -\delta\theta$. On constant variation directions h_1 and h_2, this means

$$\omega(h_1, h_2) = \omega^\mu(h_1, h_2)\mathrm{d}\Sigma_\mu, \qquad \omega^\mu = h_1\partial^\mu h_2 - h_2\partial^\mu h_1. \tag{8.2}$$

The potential is the boundary expression in the pointwise first variation:

$$\delta(L\mathrm{vol})(h) = E[\phi]h\,\mathrm{vol} + d\theta_\phi(h).$$

This identity follows from the product rule, as in Lecture 1. Its validity does not require extending that lecture's torus boundary theorem to other regions.

To check the minus sign in the current, vary the coefficient of $\theta_\phi(h_2)$ along h_1 and subtract the reverse variation:

$$(\delta\theta)(h_1, h_2) = \big[(\partial^\mu h_1)h_2 - (\partial^\mu h_2)h_1\big]\mathrm{d}\Sigma_\mu = -\omega(h_1, h_2).$$

For solutions of the linearised equation, the current is closed in spacetime. The product rule and the symmetry of the metric give

$$\partial_\mu \omega^\mu = h_1\Box h_2 - h_2\Box h_1 = -m^2 h_1 h_2 + m^2 h_2 h_1 = 0, \qquad d\omega = 0. \tag{8.3}$$

The mass terms cancel. No boundary assertion has yet been used.

8.3 A graph and its orientation

Definition 8.3 (Spacelike graph and its pairing)**.** A *spacelike graph* is a surface $\Sigma_\tau = \{(\tau(x), x) : x \in \mathbb{R}^d\}$ with $\tau \in C^1(\mathbb{R}^d)$ and $|\nabla\tau| < 1$. We orient it as the boundary of its past, with the future normal first. When the current is integrable on the graph, its pairing is

$$\Omega_{\Sigma_\tau}(h_1, h_2) = \int_{\Sigma_\tau} \omega(h_1, h_2). \tag{8.4}$$

For a constant graph $\tau = t_0$, write Ω_{t_0}.

A tangent vector to the graph has the form $(\nabla\tau \cdot v, v)$. Its Minkowski square is $(\nabla\tau \cdot v)^2 - |v|^2 < 0$ when $v \neq 0$. The future unit normal and the positive induced area measure satisfy

$$N_\mu = \frac{(1, -\nabla\tau)}{\sqrt{1 - |\nabla\tau|^2}}, \qquad dS = \sqrt{1 - |\nabla\tau|^2}\,\mathrm{d}^d x, \qquad N_\mu dS = (1, -\nabla\tau)\,\mathrm{d}^d x.$$

The area factor is the square root of the determinant of $I - \nabla\tau \otimes \nabla\tau$.

The pullback can also be read directly from the differential forms. The pullback of $\mathrm{d}\Sigma_0$ is $\mathrm{d}^d x$. In $\mathrm{d}\Sigma_i$, replace $\mathrm{d}t$ by $\partial_j\tau\,\mathrm{d}x^j$. Only the term $j = i$ survives, and its sign is negative. Consequently

$$\Omega_{\Sigma_\tau}(h_1, h_2) = \int_{\mathbb{R}^d} \big[\omega^0 - \omega^i\partial_i\tau\big](\tau(x), x)\,\mathrm{d}^d x. \tag{8.5}$$

Both surfaces in a comparison carry this future orientation. As a boundary of the region between them, the lower face instead carries the opposite sign.

Theorem 8.4 (Slice independence). *For $h_1, h_2 \in \mathcal{S}_m$ and $t_0 \in \mathbb{R}$, let $\tau \in C^1(\mathbb{R}^d)$ satisfy $|\nabla\tau| < 1$ and $\tau = t_0$ outside a compact set. Then*

$$\Omega_{\Sigma_\tau}(h_1, h_2) = \Omega_{t_0}(h_1, h_2) = \int_{\mathbb{R}^d} (h_1 \partial_t h_2 - h_2 \partial_t h_1)(t_0, x)\, \mathrm{d}^d x. \tag{8.6}$$

The value is also independent of t_0. This pairing is a nondegenerate closed alternating form on the solution space. On a constant slice, with $\pi = \phi_t$, it is $\Omega = -\delta\Theta = \int \delta\phi \wedge \delta\pi$, where $\Theta = \int \pi\, \delta\phi$.

Proof. We integrate the divergence by coordinates, including when the graph crosses $t = t_0$. Since $\tau - t_0$ has compact support, τ is bounded. Choose a bounded time interval containing both slices. Finite propagation places the current's support in one spatial rectangle Q throughout that interval. Choose Q large enough also to contain the deformation of the graph.

Define the signed time integrals

$$A^i(x) = \int_{t_0}^{\tau(x)} \omega^i(t, x)\, \mathrm{d}t.$$

They are C^1 and vanish near every face of Q. The fundamental theorem and the chain rule give, with summation over i,

$$\partial_i A^i = \int_{t_0}^{\tau(x)} \partial_i \omega^i(t, x)\, \mathrm{d}t + \omega^i(\tau(x), x)\partial_i \tau.$$

Use $\partial_t \omega^0 = -\partial_i \omega^i$ and integrate in time. Rearranging yields

$$\omega^0(\tau(x), x) - \omega^i(\tau(x), x)\partial_i \tau - \omega^0(t_0, x) = -\partial_i A^i(x).$$

Fubini and coordinate FTC on Q show that the integral of the right side is zero: its opposite face values vanish. Formula (8.5) proves the equality of pairings. This is the graph form of the boundary theorem, proved here on $\mathbb{R}^d$. For a constant graph $\tau = t_1$, the same argument works because finite propagation still makes A^i vanish near the faces. Thus all constant-time pairings agree.

On the constant slice, evaluating the wedge gives $(\delta\phi \wedge \delta\pi)(h_1, h_2) = h_1 \partial_t h_2 - h_2 \partial_t h_1$. Also $-\delta(\pi\, \delta\phi) = \delta\phi \wedge \delta\pi$. The pairing is alternating. It is closed under δ because its coefficients are independent of the background solution, so their derivatives along every affine family are zero.

Finally, suppose $\Omega_{t_0}(h, k) = 0$ for every $k \in \mathcal{S}_m$. Write $f = h(t_0)$ and $g = h_t(t_0)$, both compact smooth functions. Arbitrary compact smooth data at time t_0 produce a solution in $\mathcal{S}_m$: apply Lecture 4 after translating time and Lecture 5 backwards to time zero. Choose k with data $(0, f)$. The pairing is $\int f^2 = 0$, so $f = 0$. Choose k with data $(g, 0)$. It gives $-\int g^2 = 0$, so $g = 0$. Cauchy uniqueness gives $h = 0$. This is nondegeneracy. ∎

The term *symplectic* here means a closed alternating form with this nondegeneracy property. It does not assert that every linear functional has a representing vector. For a functional with compact smooth gradients, the equation $\iota_{X_F}\Omega = \delta F$ gives $X_F = (F_\pi, -F_\phi)$ and $X_F G = \{G, F\}$, with the functional derivatives and bracket order of Lecture 7. Compactness of these gradients ensures that this vector is tangent to the solution space used here.

8.4 Two compact waves on a deformed slice

Consider $d = 1$ and $m = 0$. Choose an even smooth cutoff χ with $\chi = 1$ on $[-3, 3]$ and $\chi = 0$ outside $[-4, 4]$. Define two right-moving solutions by

$$h_1(t, x) = \chi(q), \qquad h_2(t, x) = (q + q^2)\chi(q), \qquad q = x - t.$$

Both solve $h_{tt} - h_{xx} = 0$ and have compact smooth Cauchy data. The derivatives of the cutoff cancel in the antisymmetric product:

$$W(q) = \chi(q)\big[(q + q^2)\chi(q)\big]' - (q + q^2)\chi(q)\chi'(q) = (1 + 2q)\chi(q)^2, \qquad \omega^0 = \omega^1 = -W(q).$$

Set $A = \int_{\mathbb{R}} \chi(q)^2 \, \mathrm{d}q > 0$. Evenness of χ^2 makes the term $2q\chi^2$ odd, so $\Omega_0(h_1, h_2) = -A \neq 0$.

Deform the slice by the compact graph

$$\tau(x) = \frac{2}{5} b(x/2), \qquad b(s) = \begin{cases} (1 - s^2)^4, & |s| < 1, \\ 0, & |s| \geq 1. \end{cases}$$

This graph is C^3. For $|s| \leq 1$, its slope obeys

$$|\tau'(x)| = \frac{8}{5}|s|(1 - s^2)^3 \leq \frac{8}{5}|s|(1 - s^2) \leq \frac{16}{15\sqrt{3}} < 1, \qquad s = x/2.$$

The middle bound follows by maximising $s(1 - s^2)$ on $[0, 1]$. Thus the graph is spacelike. It agrees with $t = 0$ outside $[-2, 2]$.

Figure 8.1 shows the compact comparison. During $0 \leq t \leq 1/2$, both waves vanish outside $[-4, 9/2]$. The vertical sides $x = \pm 5$ therefore carry no current.

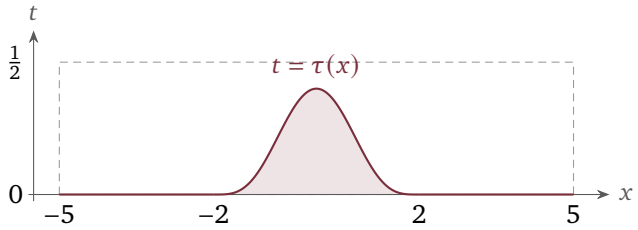


Figure 8.1: The deformed slice $t = \tau(x)$ and the constant slice $t = 0$ inside a compact comparison cylinder. The shaded region lies between the slices. The vertical sides lie outside the waves' support on the displayed interval.

The graph formula gives

$$\Omega_{\Sigma_\tau}(h_1, h_2) = -\int_{\mathbb{R}} W(x - \tau(x))(1 - \tau'(x)) \, \mathrm{d}x.$$

The change of variable $q = x - \tau(x)$ has derivative $1 - \tau'(x) > 0$ and agrees with $q = x$ outside a compact set. It is therefore a bijection of $\mathbb{R}$ onto $\mathbb{R}$. Substitution gives $\Omega_{\Sigma_\tau} = -\int W = -A$. This explicitly verifies the conserved, nonzero pairing on the two slices.

8.5 Side flux changes the pairing

Conservation of the spacetime current does not conserve its integral over a fixed spatial subregion. In one dimension, coordinate FTC gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_\alpha^\beta \omega^0(t, x) \, \mathrm{d}x = -\big[\omega^1(t, \beta) - \omega^1(t, \alpha)\big]. \tag{8.7}$$

The boundaries $x = \alpha$ and $x = \beta$ are timelike sides: they extend in the time direction at fixed spatial positions.

For an explicit local example, take $m = 0$, $h_1 = 1$ and $h_2 = (x+t)^2$. These are smooth wave solutions, although their Cauchy data are not compact. Their current and its integral on $I = [0,1]$ are

$$\omega^0 = 2(x+t), \qquad \omega^1 = -2(x+t), \qquad \Omega_I(t) = \int_0^1 \omega^0 \, \mathrm{d}x = 1 + 2t.$$

The net outward side flux is $\omega^1(t,1) - \omega^1(t,0) = -2$. Thus $\Omega_I' = 2$, in agreement with (8.7). The current is closed, but the side contribution is nonzero. Boundary conditions must control this contribution when a field is restricted to a spatial region, as in Lectures 12 and 27.

8.6 General Cauchy surfaces

A *Cauchy surface* meets every inextendible timelike curve exactly once. A timelike curve has tangent of positive Minkowski square. For the same smooth Klein–Gordon solutions with compact Cauchy data, the pairing is also finite and independent of any smooth spacelike Cauchy surface in full Minkowski spacetime, oriented by its future normal. This extension is *cited*, not proved here. It uses compactness of the solution support on general Cauchy surfaces in addition to current closure. See Khavkine, *Covariant phase space, constraints, gauge and the Peierls formula* (2014), §3.3.3, Lemma 3.11, arXiv:1402.1282. The coordinate proof above establishes the compact graph deformations needed in this course.

Summary. For smooth real Klein–Gordon solutions with compact Cauchy data, $\theta = p^\mu \delta\phi \, \mathrm{d}\Sigma_\mu$ gives $\omega = -\delta\theta$ and $d\omega = 0$. The future-oriented graph element is $(1, -\nabla\tau)\mathrm{d}^d x$. Its integral agrees with the constant-slice form $\Omega = \int \delta\phi \wedge \delta\pi$ for a compact spacelike deformation. Arbitrary compact Cauchy test data prove nondegeneracy. On a bounded spatial region, side flux can change this integral.

Self-check. For the compact waves above, reverse the order of h_1 and h_2. What is the pairing on either slice?

Answer. Antisymmetry changes $-A$ to $+A$. The order of the arguments and the sign $\omega = -\delta\theta$ must both be retained.

Self-check. On a graph $t = vx$ in one spatial dimension, with $|v| < 1$, what is the pullback density for a right-moving current with $\omega^0 = \omega^1 = -W(x-t)$?

Answer. The density is $-(1-v)W((1-v)x)\,\mathrm{d}x$. Its sign uses the future graph element $(1, -v)\mathrm{d}x$. This local formula does not require the affine graph to be a compact deformation.

Self-check. Suppose h pairs to zero with every compact-data solution. Which two choices of Cauchy test data force its displacement and velocity to vanish?

Answer. Writing the data of h as (f, g), choose test data $(0, f)$ and $(g, 0)$. The pairings are $\int f^2$ and $-\int g^2$. Their vanishing gives $f = g = 0$, and uniqueness then gives $h = 0$.

Self-check. For the interval example, compare $\Omega_I(1) - \Omega_I(0)$ with the integrated outward side flux from $t = 0$ to $t = 1$.

Answer. The change is $3 - 1 = 2$. The integrated outward flux is -2. Their sum is zero, as required by the local boundary balance.